Breaking Defensive Structures

Working Project Concept — CS715 Machine Learning

The Idea

I want to investigate how individual soccer players and their actions affect an opposing team's defensive structure, particularly when playing against compact, organized defenses such as low blocks.

Rather than focusing primarily on goals, assists, or expected goals (xG), I want to understand the actions that make those opportunities possible in the first place.

For example, what actions help spread defenders apart, create gaps between defensive lines, pull defenders out of position, or create space for teammates? How do the locations and timing of these actions affect their success? And do certain players consistently perform better in these situations?

Four Types of Defensive Disruption

1. **Penetration:** Moving the ball or a receiving player beyond defensive lines.
2. **Stretching:** Pulling defenders wider, deeper, or farther forward to reduce compactness.
3. **Space creation:** Opening passing lanes, gaps, or opportunities for teammates.
4. **Disorganization:** Creating changes in defensive positioning that make the defensive structure easier to exploit.

Central Research Question

How do individual player actions, positioning, defensive configurations, and preceding attacking sequences relate to the disruption of compact defensive structures in international soccer?

Machine Learning Approach

I plan to compare three progressively more informative models:

- **Model 1 — Event-based:** Uses individual actions, ball locations, and basic event information.
- **Model 2 — Spatial:** Adds the visible positions and configurations of opposing defenders.
- **Model 3 — Sequence-aware:** Adds preceding actions, ball movement, and attacking buildup patterns.

The goal is to determine whether additional spatial and sequential information helps identify and predict effective attacking actions.

Data and Scope

The primary data will come from StatsBomb's publicly available international tournament datasets, including Euro 2020, World Cup 2022, and Euro 2024.

The analysis will focus on international soccer rather than club competitions. Because the available positional data consists of snapshots rather than continuous tracking, the project will use measurable indicators of defensive disruption instead of attempting to track every player's movement.

Intended Outcome

I want to identify which types of actions and attacking patterns are associated with disrupting compact defenses, whether defensive positioning and buildup context improve predictions, and what these findings might suggest about different player tendencies and effectiveness.